

Social Network Aggregation

Yutong Yi; Weiwei Chen

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Institute of Logic and Cognition and Department of philosophy,
Sun Yat-sen University, China

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Introduction

Limitations of Single-View and the Wisdom of "Puzzling Together"

Complexity of Social Networks:

- Diverse relationships between users (friends, followers, interests, etc.).
- A single perspective (e.g., only "friends") is like looking through a keyhole, providing partial information.

A Vivid Analogy: Bałtyk Tower



Limitations of a single perspective.
Look like a normal office building

Front View



Back View



Side View



Upward View

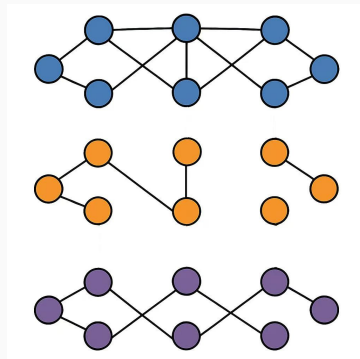


Core Idea: Fusing information from "photos" (perspectives) taken from different angles leads to a more **complete**, **multi-dimensional**, and **accurate** understanding than any single photo.

“Perspectives” in Social Networks and Their Aggregation Value

Some Examples of Perspectives in Social Networks Include:

- **Relationship Perspective:** Friend, Classmate, relatives etc.
- **Behavioral Perspective:** Like, comment, repose interaction etc.
- **Content Perspective:** social networks based on similarity of posted content (text, images) etc.
- **Attribute Perspective:** Graphs based on shared user attributes (e.g., interest tags, geographical location etc.)



(Multi-layer networks represent different perspectives)

Why Combine Different Views of Social Networks?

- **Get Smarter Suggestions for Content** We can imagine that the content and friends of the same people are really different on LinkedIn, Trip.com, and X. The aggregation of social networks could give more creative suggestions.
- **A Fuller Picture of People's Attitudes Towards Social Issues:** Towards the same issue, people may have different positions in different aspects. The aggregation helps us to get a more accurate overview of people's attributions.
- **Weaken the Majority Illusion** In a single social network, people are easily affected by the majority illusion and have an illusory judgment about the full view. With multi-view social networks, it gives more chances to connect with different information.

Conclusion: Studying how to combine different social network views helps us understand tricky social situations more deeply and show the possibility of further applications.

Preliminary

A social network is made up of actors and relationships.

Fix the set of actors A . A social network $G = \langle A, E \rangle$, based on A , is defined by a set of edges $E \subseteq A \times A$ and can be seen as a undirected simple graph.

Everyone in a finite group of individuals $\mathcal{N} = \{1, \dots, n\}$ provides a social network, giving rise to a profile $\mathbf{E} = (E_1, \dots, E_n)$.

An aggregate rule is a function mapping profile to collective graphs

$$F : (2^{A \times A})^n \longrightarrow 2^{A \times A}$$

We may want to impose certain axioms on F , e.g.:

- **Anonymous:** $F(E_1, \dots, E_n) = F(E_{\sigma(1)}, \dots, E_{\sigma(n)})$
- **Unanimous:** $F(\mathbf{E}) \supseteq E_1 \cap \dots \cap E_n$
- **Grounded:** $F(\mathbf{E}) \subseteq E_1 \cup \dots \cup E_n$
- **Neutral:** $N_e^{\mathbf{E}} = N_{e'}^{\mathbf{E}}$ implies $e \in F(\mathbf{E}) \iff e' \in F(\mathbf{E})$
- **Independent:** $N_e^{\mathbf{E}} = N_e^{\mathbf{E}'}$ implies $e \in F(\mathbf{E}) \iff e \in F(\mathbf{E}')$

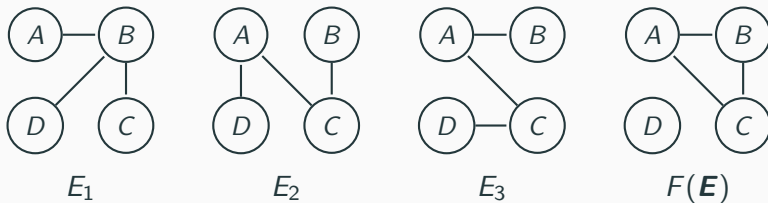
Here are several aggregation rules considered in this paper.

- **Majority rule:** $F_{maj} : \mathbf{E} \mapsto \{e \in A \times A : |N_e^{\mathbf{E}}| \geq \frac{n}{2}\}$
- **Nomination rule:** $F_{\cup} : \mathbf{E} \mapsto \bigcup_{i=1}^n \mathbf{E}_i.$
- **Dictatorial rule:** $\exists i \in \mathcal{N}$ s.t. $F_i : \mathbf{E} \mapsto E_i.$
- **Oligarchy rule:** $\exists C \subseteq \mathcal{N}, C \neq \emptyset$ s.t. $F : \mathbf{E} \mapsto \bigcap_{i \in C} E_i.$

Aggregation Rule

F preserve δ : $\forall E_i \in \mathbf{E} \in \delta \rightarrow F(\mathbf{E}) \in \delta. (\delta \text{ is a semantic property.})$

Example

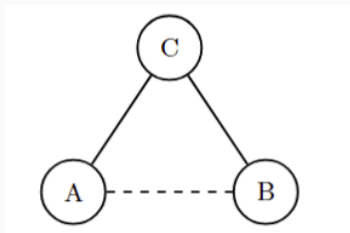


The majority rule does not preserve connectivity.

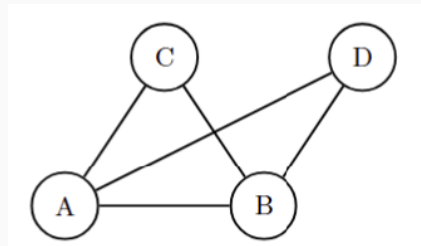
Properties

Transitivity and Connectivity

- **Transitivity:** $(a, c), (b, c) \in E$
implies $(a, b) \in E$.

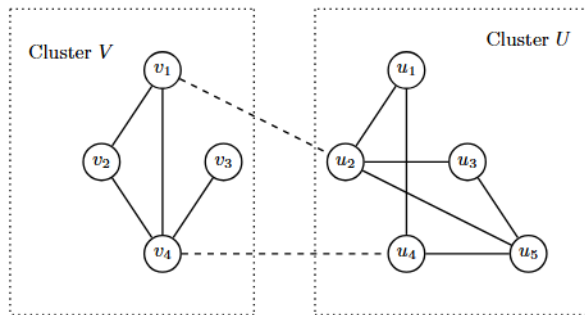


- **Connectivity:**
 $\forall a, b \in A, \exists P(a, b) \subseteq E$.



Weak Tie

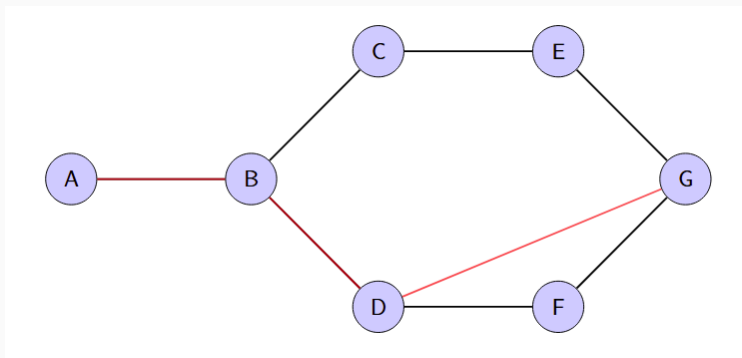
- **Cluster:** V is a cluster if the subgraph $G_V = \langle V, E_V \rangle$ induced by set $V \subseteq A$ is a connected graph and $|E_V| \geq |V|$.
- **Weak tie:** We say there are **weak ties** between two distinct clusters V and U when $1 \leq |E(V, U)| \leq 2$, where $E(V, U) = \{(v_i, u_j) | v_i \in V, u_j \in U \text{ and } (v_i, u_j) \text{ is a edge in the social network}\}$.



Six Degree of Separation

- Six degree of separation: ¹

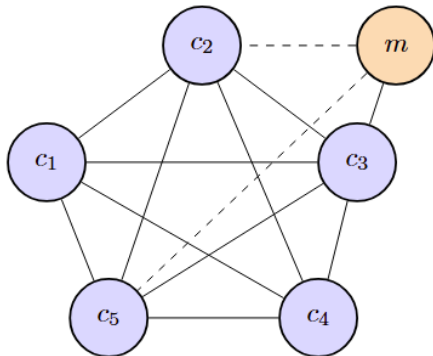
$$\forall a, b \in A \quad |P(a, b)| \leq 6$$



¹At this point $P(a, b)$ denotes the shortest path from a to b

Margin

- **Margin Actor:** Given a clique $C = \{c_1, \dots, c_n\} \subseteq A$, an actor $a \in A \setminus C$ is a **margin actor** with respect to C if $\exists c_i \in C$ such that $(a, c_i) \in E$.
- **Margin Property:** A graph $G = (A, E)$ satisfies the **margin** property if there exists a clique $C \subseteq A$ such that $\exists a \in A \setminus C$ with $\exists c_i \in C, (a, c_i) \in E$.



Result

Fact 1: For $|\mathcal{N}| > 3$, any unanimous, grounded, and independent aggregation rule F preserves transitivity, connectivity, margin, weak tie is neutral.

Explanation: This fact shows that when **basic fairness conditions** are satisfied, many important structural properties of social networks **could be preserved** during aggregation. These include how influence or relationships pass through the network (transitivity), how strongly actors are connected (margin), and whether the structure remains balanced (connectivity).

Proposition 1: For $|\mathcal{N}| \geq 3$, when the number of networks being aggregated is odd the majority rule F always preserves the property of margin of a fixed 2-clique.

Explanation: This proposition show that the majority rule performs particularly well **in small, tightly connected substructures**. For instance, in small cliques, it's not hard to preserve the connections of key local features with others.

Theorem 1 For $|\mathcal{N}| \geq 3$, the nomination rule F preserves connectivity.

Theorem 2 For $|\mathcal{N}| \geq 3$, the nomination rule F preserves six degree of separation.

Explanation: These results highlight the strength of the nomination rule in **preserving global and structural features** of social networks. It ensures that the network remains connected and six degree of separation support **robust communication across sub-social-networks**.

Theorem 3 Any unanimous, grounded and independent rule preserves transitivity for some social networks must be oligarchy.

Explanation: To always preserve these transitivity properties, the aggregation rule must allow decisions to be **dominated by a small group** (oligarchy).

Theorem 4 Any unanimous, grounded and independent rule preserves margin and weak tie for some social networks must be dictatorial.

Explanation: Preserving all these properties simultaneously forces the rule to be dictatorial—completely **determined by a single individual**—thus sacrificing fairness.

Conclusion

This paper introduces a formal model for aggregating social networks, enabling us to analyze how different structures merge. Inspired by actual social interactions and social network studies in social science, we formally define desirable properties and investigate how various aggregation rules preserve them. Ultimately, we report our findings on which properties are preserved under specific aggregation rules.

to be continued...

- Using hypergraphs instead of simple graphs to indicate relationships among more than two people.
- The dynamic process of changes in social networks.
- The bipolar relationships between actors in social networks (such as 'follow' vs. 'block')
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